Assignment 3

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Aim
To design and develop a dynamic web application similar to a fashion store or e-commerce website using HTML, CSS, and JavaScript, with the option of using Bootstrap CDN for better design and responsiveness.

Objective
To create and implement an interactive e-commerce web application using HTML, CSS, and JavaScript, enhanced with Bootstrap to achieve a modern, responsive, and user-friendly interface.

Theory
This project focuses on building a dynamic e-commerce website using HTML, CSS, JavaScript, and Bootstrap. Each technology has a specific role in creating a responsive and interactive shopping experience.

1. HTML (HyperText Markup Language)
HTML provides the basic structure of the web application. It organizes content using elements like <nav>, <header>, <main>, <section>, and <footer>. These elements help structure navigation menus, product displays, and shopping cart details. Using semantic HTML improves accessibility, search engine optimization (SEO), and code readability.

2. CSS (Cascading Style Sheets)
CSS is responsible for styling the application and making it visually attractive. It controls colors, fonts, layouts, and spacing. CSS variables (defined using :root) help manage themes easily. Media queries are used to make the design responsive across different devices. Effects like hover animations and transitions improve user interaction, while focus states enhance accessibility.

3. Bootstrap Framework
Bootstrap is a popular CSS framework that speeds up development by providing ready-made components and a grid system. By using Bootstrap CDN, the project benefits from responsive layouts, navigation bars, buttons, cards, and more. It ensures consistency and saves development time.

4. JavaScript (JS)
JavaScript adds interactivity and dynamic behavior to the application. It updates the webpage content using DOM manipulation and manages features like adding items to the cart, removing

items, updating quantities, and clearing the cart. Functions such as addToCart, removeItem, updateQuantity, and clearCart are used. It also handles validation and error checking to ensure smooth functioning.

5. Local Storage
The browser's localStorage is used to store cart data permanently. This allows users to keep their selected items even after refreshing or closing the browser. Data is stored in JSON format, making it easy to retrieve and update.

6. Responsive Design
Responsive design ensures that the application works well on all screen sizes, including mobiles, tablets, and desktops. This is achieved using CSS media queries and Bootstrap's grid system to adjust layouts automatically.

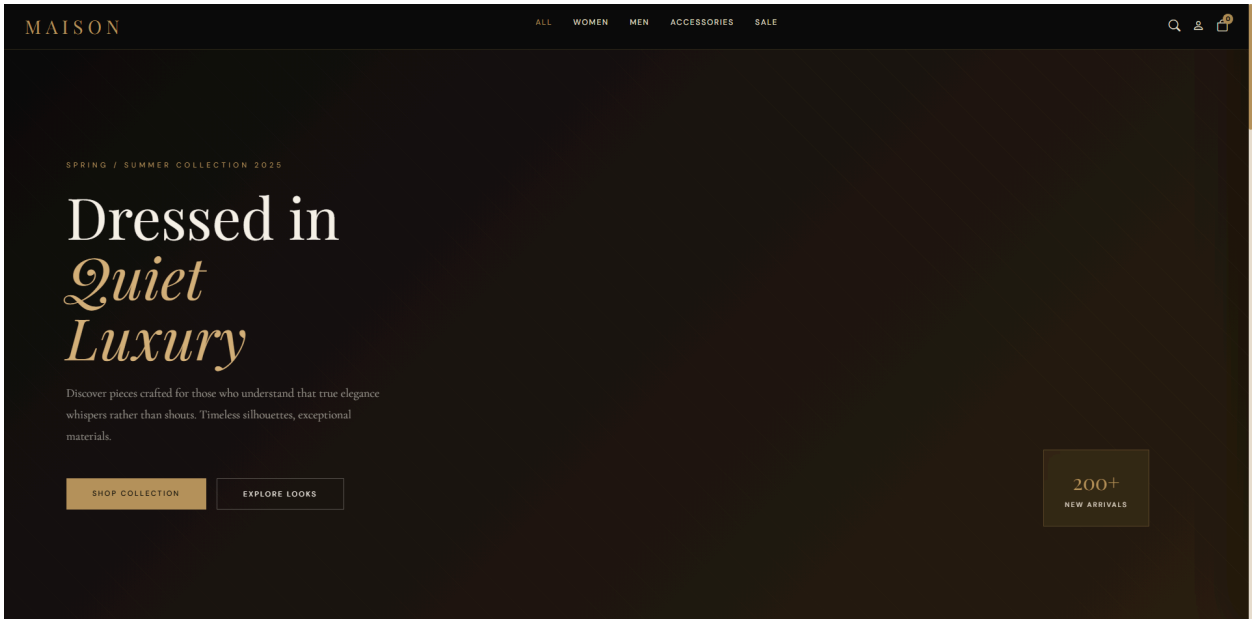
7. Accessibility
Accessibility features make the app usable for all users, including those with disabilities. This includes using semantic HTML, adding alt text to images, maintaining proper focus states, and ensuring good color contrast.

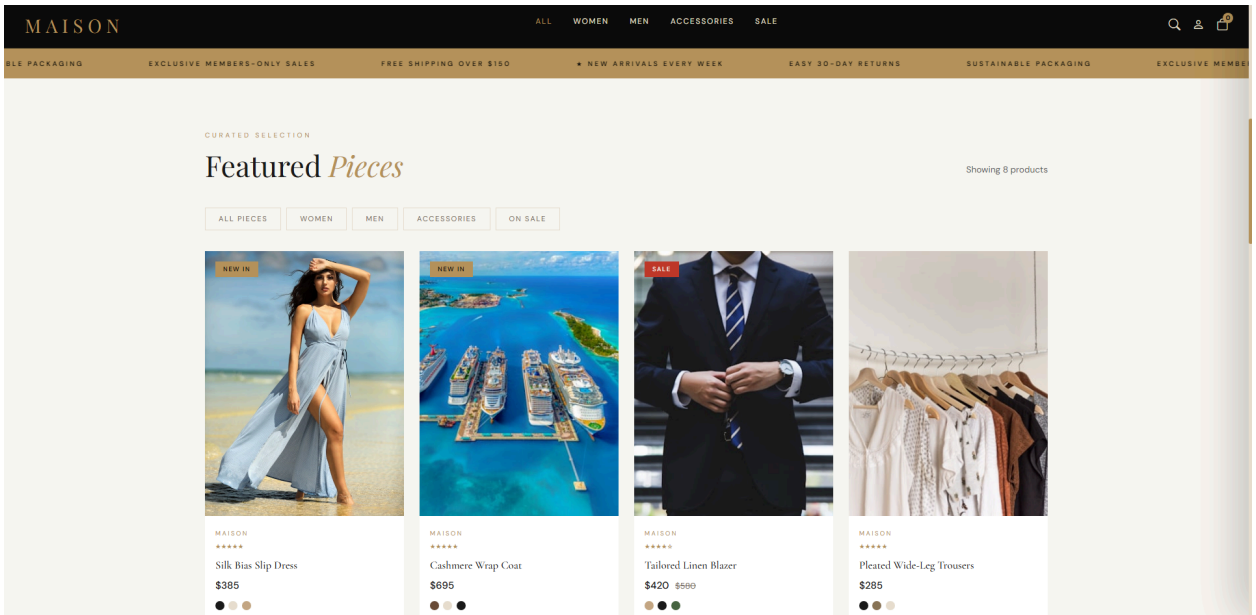
8. User Experience (UX) Enhancements
Features like notifications, confirmation messages, and smooth animations improve the user experience. Interactive elements such as quantity controls and automatic price updates make the shopping process easier and more engaging.

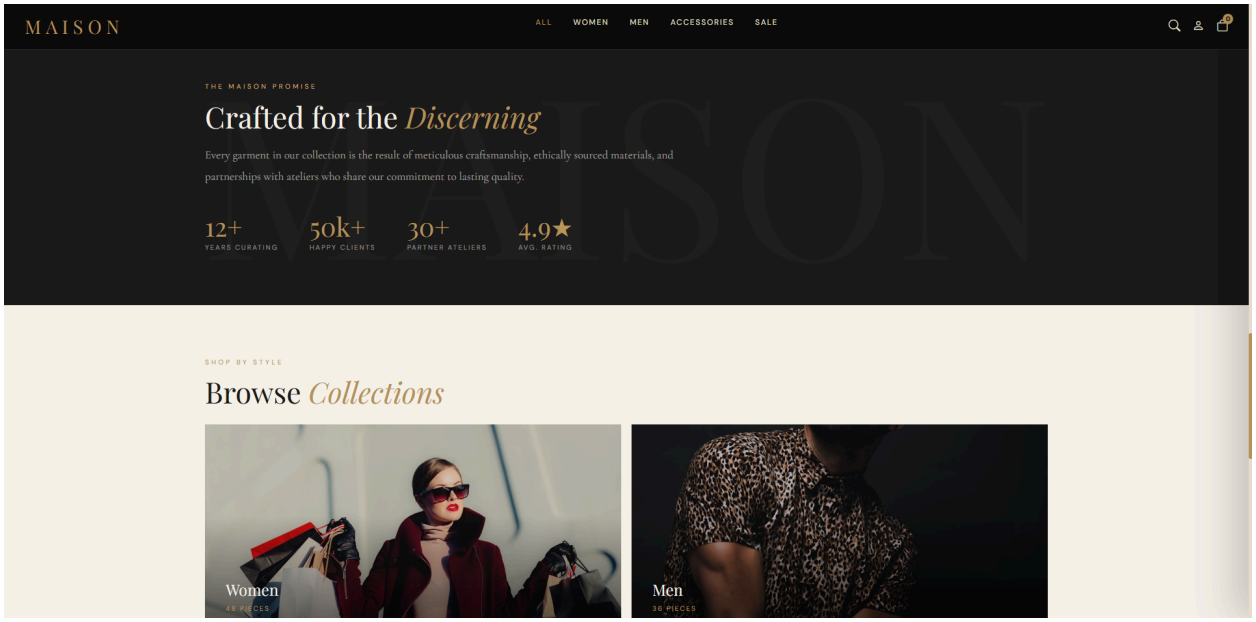
9. Code Organization and Maintainability
The project follows a structured approach by separating HTML, CSS, and JavaScript into different files. This makes the code easier to manage, update, and scale in the future.

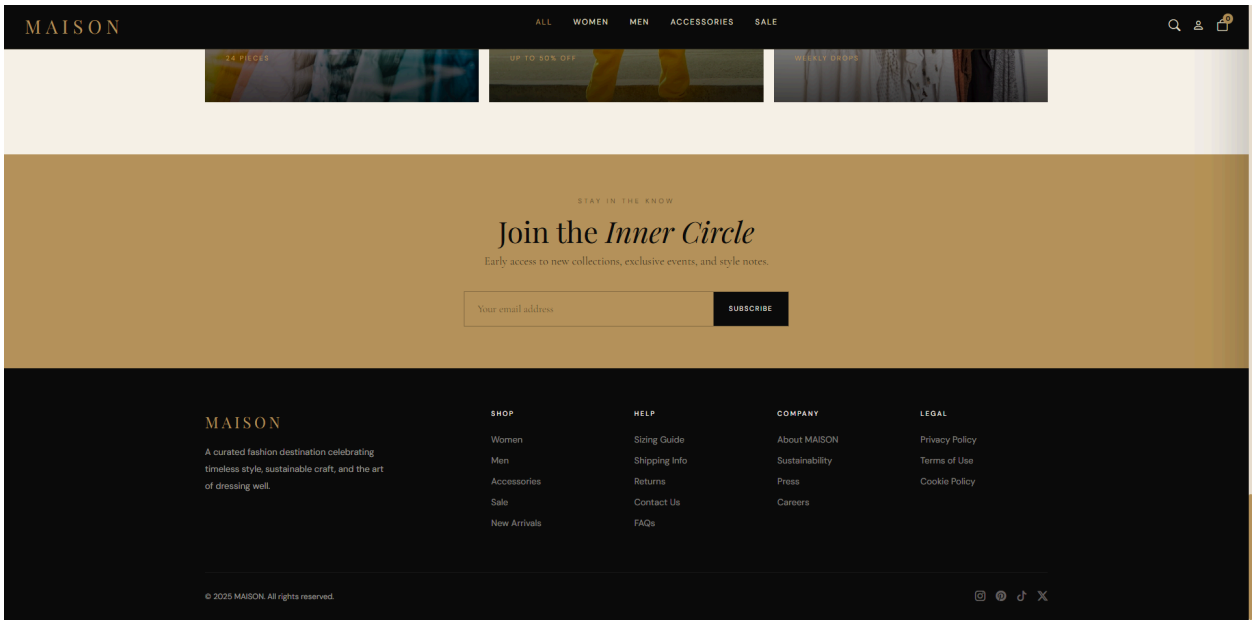
10. Modern Web Practices
The application uses modern development techniques such as ES6+ JavaScript, CSS variables, and Bootstrap. These practices ensure better performance, cleaner code, and a professional user interface.

Output -


MAISON
ALL
WOMEN
MEN
ACCESSORIES
SALE
SPRING / SUMMER COLLECTION 2025
Dressed in
Quiet
Luxury
Discover pieces crafted for those who understand that true elegance whispers rather than shouts. Timeless silhouettes, exceptional materials.
SHOP COLLECTION
EXPLORE LOOKS
200+
NEW ARRIVALS


MAISON
ALL
WOMEN
MEN
ACCESSORIES
SALE
EXCLUSIVE MEMBERS-ONLY SALES
FREE SHIPPING OVER $150
NEW ARRIVALS EVERY WEEK
EASY 30-DAY RETURNS
SUSTAINABLE PACKAGING
CURATED SELECTION
Featured Pieces
Showing 8 products
ALL PIECES
WOMEN
MEN
ACCESSORIES
ON SALE
NEW IN
NEW IN
SALE
MAISON
Silk Bias Slip Dress
$385
MAISON
Cashmere Wrap Coat
$695
MAISON
Tailored Linen Blazer
$420 $580
MAISON
Pleated Wide-Leg Trousers
$285





Conclusion:

The project successfully demonstrates how to build a visually appealing and interactive fashion store web app using core web technologies and Bootstrap. It highlights the importance of responsive design, user-friendly interfaces, and dynamic features in modern e-commerce solutions.